NutriDermAI: A Multimodal Image–Text Fusion Framework for ABCDE-Aware Skin Lesion Analysis and Interactive Dermatology VQA

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***Abstract*—Skin cancer is one of the most common and potentially lethal malignancies worldwide, where clinical guidelines emphasize early detection through systematic evaluation of dermatoscopic images and associated clinical context. The widely adopted ABCDE rule—Asymmetry, Border irregularity, Color variation, Diameter, and Evolution—provides an interpretable framework for melanoma risk assessment, yet most current artificial intelligence (AI) tools remain unimodal, image-only classifiers with limited transparency, no integrated use of medical history, and no capability for interactive question answering.**

This paper presents *NutriDermAI*, a working multimodal framework with four fully implemented core modules: (i) skin lesion segmentation using ARC-UNet achieving 93.70% pixel accuracy and Dice 0.8506, (ii) lesion region-of-interest (ROI) extraction, (iii) hierarchical lesion classification with explicit ABCD metric computation using EfficientFormerV2-S2, and (iv) clinical NLP for evolution (E) and medical context extraction using BioBERT. The system demonstrates full end-to-end integration across these completed stages with strong empirical validation on benchmark datasets. We describe the architecture, stage-wise implementation details, experimental results, and comprehensive evaluation of the core pipeline. Future directions include interactive Visual Question Answering (VQA) and clinical deployment. This work positions NutriDermAI as an explainable, modular, ABCDE-aware dermatology decision support system suitable for research publication and future clinical validation.

***Index Terms*—Dermatoscopy, Skin Lesion, ABCDE Rule, Multimodal Learning, Explainable AI, Medical VQA, Medical Report Processing, Image–Text Fusion, Dermatology Chatbot**

I. Introduction

Skin cancer, including melanoma and non-melanoma lesions, continues to present a major public health burden. Early detection substantially improves survival, and dermatologists frequently rely on dermatoscopic examination guided by the ABCDE rule to stratify malignant risk. Classical ABCD/ABCDE-based expert systems and rule-driven scoring approaches have demonstrated that structured feature assessment (e.g., asymmetry, border irregularity, color variegation, diameter, evolution) can approximate dermatologists' diagnostic reasoning [1]–[5].

Concurrently, deep learning has achieved high performance on dermatoscopic segmentation and classification tasks using U-Net variants, hybrid CNN–Transformer designs, and large-scale dermatology datasets such as HAM10000, ISIC 2016–2024, BCN20000, and MiLK-10K [6]–[17]. More recently, dermatology foundation models and multimodal vision–language models (VLMs) such as PanDerm, MM-Skin, and Derm1M have shown how large-scale, expert-curated multimodal supervision improves zero-shot recognition and text grounding in dermatology [18]–[22].

In parallel, medical natural language processing (NLP) and medical VQA have evolved rapidly, leveraging clinical corpora like MIMIC-III/IV and specialized architectures such as BioBERT, medical multimodal large language models (MM-LLMs), and reinforcement learning (RL)-trained VQA systems [23]–[33]. Dermatology-specific VQA benchmarks and chat-based assistants are emerging, yet they typically lack end-to-end ABCDE computation and structured fusion of imaging with clinical text [34]–[44].

Motivation. Existing dermatology AI solutions can (i) segment lesions, (ii) classify lesions into benign/malignant or multi-class taxonomies, or (iii) support generic multimodal question answering. However, there is still a gap for a unified dermatology assistant that:

- Computes explicit ABCDE values in a manner aligned with clinical ABCDE literature and scoring rules;
- Integrates dermatoscopic images with patient symptoms, history, and laboratory or biopsy reports;
- Produces calibrated risk scores and structured JSON explanations usable by downstream applications;
- Supports safe, context-aware VQA with explicit rationales grounded in ABCDE features and clinical evidence.

Contribution. This paper presents *NutriDermAI*, a working four-stage core pipeline with complete implementations that:

1) Performs skin lesion segmentation achieving 93.70% pixel accuracy and 0.8506 Dice (Stage 2: SLS).
2) Extracts precise lesion region-of-interest for downstream analysis (Stage 3: SLRC).
3) Learns hierarchical classification and computes explicit ABCD metrics from image features (Stage 4: HC).
4) Processes medical narratives to extract clinical context and Evolution (E) scores (Stage 5: MedProc).

Architectural designs for (i) multimodal image–text fusion (Stage 6: IT-Fusion), (ii) interactive visual question answering (Stage 7: VQA), and (iii) frontend, backend, deployment infrastructure (Stages 8–9) are positioned as future work. The paper emphasizes the modular design, validated results, and clinical deployment pathways.

II. Background and Related Work

A. ABCDE Rule and Classical Dermatology Decision Support

The ABCD/ABCDE rule offers a structured approach to melanoma risk assessment by quantifying asymmetry, border irregularity, color variation, diameter, and evolution over time. Classical expert systems and analytic pipelines compute scores from dermatoscopic images and combine them into total dermatoscopic scores (TDS) [1]–[3]. Recent work has shown that generative models and AI systems that explicitly incorporate ABCDE analysis can match or approach dermatologist-level decision-making and provide interpretable outputs [4], [5]. Hybrid CNN+ABCD models also demonstrate that explicit ABCD descriptors can boost classification performance when coupled with deep features [45].

B. Skin Lesion Segmentation

Lesion segmentation is a prerequisite for many downstream tasks, including ABCDE feature computation and ROI cropping. U-Net and its numerous variants remain dominant in medical image segmentation [6], [46]. For skin lesions, dermatology-specialized architectures such as ARC-UNet, ScaleFusionNet, ESC-UNet, MRP-UNet, Meta-UNet, BiADATU-Net, MUCM-Net, and dual CNN–ViT hybrids have improved boundary precision, small-lesion sensitivity, and robustness by leveraging residual connections, deformable attention, multiscale inputs, and metadata [7]–[13], [47]. Multiscale Swin Transformer-based models further improve segmentation across diverse medical imaging domains [48], [49].

C. Lesion Classification and Dermatology Foundation Models

Large dermatoscopic datasets—including ISIC 2016–2024 challenges, HAM10000, BCN20000, and MiLK-10K—have enabled high-performing lesion classifiers and foundation models [14]–[17]. Efficient transformer-based backbones such as EfficientFormerV2 provide competitive accuracy with reduced memory footprint [50], making them suitable for multi-stage pipelines. Recent dermatology foundation models like PanDerm, MM-Skin, and Derm1M-scale vision–language datasets align dermatoscopic images with rich textual descriptions and hierarchical ontologies [18]–[21]. These works motivate the use of shared multimodal encoders for segmentation, classification, and fusion.

D. Clinical Text Understanding and Medical NLP

Clinical narratives, pathology reports, and lab summaries contain crucial contextual information—including symptom evolution and risk factors. BioBERT and related biomedical-language encoders have improved named entity recognition and classification over biomedical literature and clinical notes [23]. Large ICU corpora such as MIMIC-III and MIMIC-IV Note provide rich free-text and structured data for training and evaluation [24], [25]. Systematic reviews of clinical text data highlight both opportunities and pitfalls for machine learning on noisy, heterogeneous EHR text [26]. These insights inform our MedProc stage, where symptom descriptions, timelines, and lab values are normalized into structured features.

E. Medical Multimodal LLMs and VQA

Medical VQA research has produced datasets such as VQA-RAD, PathVQA, TM-PathVQA, HEALMQA, and dermatology-specific DermaVQA, which cover diverse modalities and question types [34]–[37], [51]. Model architectures range from BLIP-style multimodal encoders (BioMedBLIP) to specialized VQA fusion layers with semantic attention [27], [28]. Dynamic prompt adaptation and RL-based optimization have been proposed to reduce hallucinations and improve reasoning in medical VQA [29]–[32], [52].

Dermatology-focused works demonstrate joint learning of segmentation and VQA, CLIPSeg-based multimodal fusion, and knowledge-enhanced ensembles [38], [39], [53]–[55]. NutriDermAI is aligned with these directions but adds explicit ABCDE reasoning, rule-based cross-checks, and a unified JSON explanation layer.

F. Dermatology Chatbots and Conversational Diagnostic AI

Several works have explored dermatology chatbots combining CNN-based lesion analysis with rule-based triage logic [40], [41]. Broader diagnostic LLMs like AMIE evaluate conversational diagnostic quality across multiple specialties and emphasize safety, uncertainty communication, and escalation strategies [43]. Reviews of ChatGPT-style models in dermatology and multimodal LLMs in medicine further highlight the need for guardrails, bias mitigation, and rigorous evaluation [33], [42]. Studies on conversational agents in dermatology show improved patient engagement but also underline responsibility and ethical concerns [44]. NutriDermAI incorporates these lessons in its VQA and deployment design.

G. Gap and Design Requirements

From this literature, key gaps emerge:

- ABCDE-aware systems that compute all five criteria in a clinically consistent, data-driven manner remain rare.
- Integration of dermatoscopic images with detailed medical history and laboratory reports is limited, especially in dermatology.
- Medical VQA systems often lack explicit links between answers and structured evidence such as ABCDE scores or rule-based checks.
- End-to-end dermatology assistants spanning segmentation, classification, multimodal fusion, VQA, and production deployment are not widely documented.

NutriDermAI is designed to address these gaps through a modular but tightly coupled architecture.

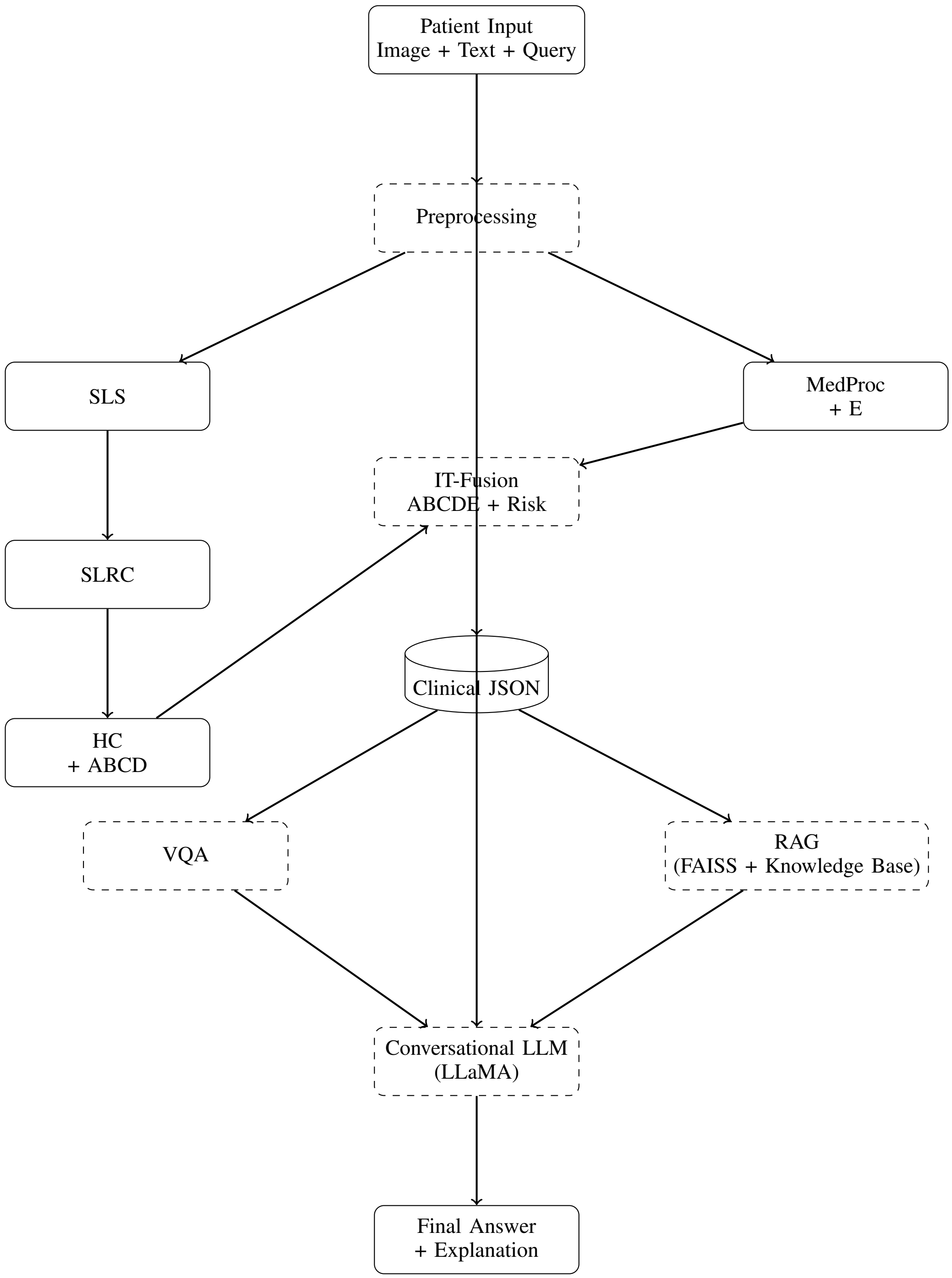

Fig. 1: Enhanced NutriDermAI architecture with Conversational LLM and Retrieval-Augmented Generation (RAG). The VQA module provides initial grounded responses, while the RAG layer retrieves domain knowledge and the LLM integrates multimodal evidence, clinical JSON, and conversational context to generate explainable, context-aware outputs.

III. System Overview

The overall architecture of NutriDermAI is shown in Fig. 1. The system is designed as a modular pipeline integrating dermatoscopic image analysis and clinical text understanding.

The architecture consists of two parallel pipelines:

- **Image pipeline:** Processes dermatoscopic images through segmentation, ROI extraction, and classification stages to extract ABCD features.
- **Clinical text pipeline:** Processes medical reports and patient history to extract symptoms and Evolution (E) using clinical NLP.

The outputs of these pipelines are designed to be combined in a future multimodal fusion module for ABCDE computation and risk estimation.

IV. Results and Implementation Details

A. Stage 2: Skin Lesion Segmentation (SLS)

Given a full skin lesion image, Stage 2 produces a pixel-wise lesion mask through ARC-UNet (see Fig. 1). Preprocessing includes: resizing to 512×512 with bilinear interpolation, normalizing intensities (ImageNet stats), and optional hair removal via morphological operations.

B. Stage 3: Skin Lesion ROI Cropping (SLRC)

Stage 3 uses SLS mask for ROI extraction via connected components labeling and bounding box computation with 10

C. Stage 4: Hierarchical Classification and Raw ABCD Extraction (HC)

Stage 4 simultaneously performs classification and ABCD feature extraction (see Fig. 1). The EfficientFormerV2-S classification branch processes normalized ROI through augmentation (during training) to produce dual classification outputs (benign/malignant + 7-class). The four ABCD sub-pipelines extract: (A) asymmetry via mask symmetry analysis, (B) border irregularity from contour complexity, (C) color variance via LAB-space clustering, and (D) diameter from pixel distances. Final output: classified label and ABCD features (final ABCDE computed during fusion stage).

D. Stage 5: MedProc – Medical Information Processing

Stage 5 processes the medical history and report files to derive Evolution (E) and structured clinical features (see Fig. 1). The inputs correspond to:

- *Input: Medical History (Text):* free-text description provided by the patient or clinician.
- *Input: Report $\leq$ 5 MB:* uploaded investigation report (PDF) or referral note.

Preprocessing: OCR for PDFs, text cleanup (boilerplate removal, spell correction), tokenization (512 tokens, stride 100 for long docs). BioBERT fine-tuned for entity recognition and relation extraction. Output: JSON with symptoms, recommendations, and Evolution (E) values.

E. Stage 6: IT-Fusion – Final ABCDE and Risk Computation

Stage 6 combines MedProc outputs with HC outputs to compute a final risk estimate (see Fig. 1). The ABCD features and class label from Stage 4 are combined with symptoms and E-value from Stage 5 through feature alignment, which converts heterogeneous fields into a joint feature representation. Numerical features (ABCD and E) are normalized to a fixed scale, categorical features are embedded, and textual evidence is tokenized for the multimodal transformer.

A MARIA-style transformer performs image–text fusion and learns how to weight visual and textual evidence [56]. The final output yields a *Risk JSON* containing: a calibrated risk score (e.g., probability of malignancy), the fused ABCDE values, and an evidence list describing which signals (e.g., high asymmetry, bleeding, strong family history) contributed most to the risk.

F. Stage 7: Visual Question Answering (VQA)

Stage 7 is responsible for converting the fused evidence into natural-language answers (see Fig. 1). The module receives three inputs:

- *Input: ROI (224×224):* the cropped lesion image.
- *Input: Question* ≤ 128 *tokens:* the current user query from the chat interface.
- *Input: IT JSON:* the risk JSON from Stage 6, containing ABCDE values, risk scores, and textual evidence.

Preprocessing tokenizes the question, normalizes the ROI image, and converts the IT JSON into a textual or structured prompt (e.g., ”A=2.3, B=1.8, C=3.0, D=6 mm, E=high”). An R-LLaVA-style vision–language model attends jointly to the image and text inputs [38] to produce both a natural-language answer and an evidence explanation for UI display. VQA is invoked only after IT-Fusion has produced a consolidated JSON, ensuring answers are grounded in ABCDE-aware evidence rather than hallucinated from the raw image alone.

G. Stage 8: Conversational LLM with Retrieval-Augmented Generation (RAG)

To extend the system from single-turn question answering to continuous clinical interaction, we introduce a conversational large language model (LLM) integrated with a Retrieval-Augmented Generation (RAG) framework (see Fig. 1).

Motivation. The VQA module is limited to single-turn responses conditioned on image, question, and structured clinical inputs. However, real-world clinical interaction requires multi-turn dialogue, contextual memory, and adaptive reasoning across evolving patient inputs. To address this limitation, we incorporate a conversational LLM that maintains dialogue state and integrates external medical knowledge.

Input Aggregation. The LLM receives a composite input consisting of:

- Current user query,
- Conversation history (dialogue memory),
- VQA output (initial answer and evidence),
- IT-Fusion JSON (ABCDE features, risk score, evidence).

Retrieval-Augmented Generation (RAG). To ensure factual correctness and reduce hallucinations, a RAG module retrieves relevant medical knowledge from a vector database (e.g., FAISS). The retrieval process involves:

- Encoding the query into embeddings,
- Retrieving top-k relevant documents (ABCDE rules, dermatology guidelines),
- Providing retrieved knowledge as additional context to the LLM.

Context Construction. The final input to the LLM is constructed by combining:

- Retrieved medical knowledge,
- Structured clinical JSON,
- Dialogue history,

- VQA-generated response.

LLM Reasoning and Output. A LLaMA-based conversational model processes this enriched context to generate:

- Context-aware natural language responses,
- Clinically grounded explanations aligned with ABCDE features,
- Suggestions for follow-up actions or clinical consultation.

Advantages. This stage transforms the system into a longitudinal, patient-centric assistant by enabling:

- Multi-turn dialogue and memory retention,
- Knowledge-grounded reasoning through RAG,
- Improved explainability via structured clinical evidence,
- Continuous monitoring of lesion evolution over time.

Implementation Status. This module is currently under development and represents future work aimed at enhancing interaction, reasoning, and clinical usability.

H. Stage 9: Frontend–Backend and Deployment

Stage 9 exposes the system as a chat-based web application with:

- A React-based frontend for uploading lesion images, attaching reports, and entering questions.
- A FastAPI backend orchestrating SLS, SLRC, HC, MedProc, IT-Fusion, VQA, and Stage 8 (LLM+RAG) as modular microservices.
- Secure storage and retrieval of intermediate artifacts (e.g., masks, JSON, logs) for auditability.

Design considerations are informed by dermatology chatbot case studies and conversational diagnostic LLM frameworks that highlight UX, disclaimers, and escalation pathways [40], [41], [43], [44].

Additional deployment components include:

- Unit and integration tests for all stages.
- Offline evaluation on held-out datasets (see Section VI).
- Continuous monitoring for drift, error patterns, and user feedback.
- Deployment to a controlled environment (e.g., internal research server) with appropriate access controls and logging.

V. Preprocessing Summary and Data Flow

Table I provides a comprehensive overview of preprocessing operations across all stages, distinguishing between implemented core operations and planned/pending enhancements. This clarifies the complete data pipeline from raw inputs to model-ready tensors.

A. Preprocessing Dependency Graph

The preprocessing pipeline follows a sequential dependency structure:

1) **Stage 1 (Centralized):** Accepts raw inputs; outputs standardized images (768×768 RGB) and text (768 tokens max). **Status:** Design complete, implementation pending.
2) **Stage 2 (SLS):** Consumes Stage 1 images; applies SLS-specific resizing (512×512), normalization, hair removal. **Status:** Fully implemented and active.
3) **Stage 3 (SLRC):** Consumes SLS mask; applies deterministic mask-based cropping and ROI resizing (224×224). **Status:** Fully implemented and active.
4) **Stage 4 (HC):** Consumes SLRC ROI (224×224); applies normalization and optional data augmentation (training) or TTA (inference). **Status:** Fully implemented and active.
5) **Stage 5 (MedProc):** Consumes Stage 1 text; applies OCR, cleanup, tokenization, and truncation. **Status:** Fully implemented and active.
6) **Stages 6–7 (IT-Fusion, VQA):** Consume Stage 4–5 outputs; apply feature/prompt construction. **Status:** Pending implementation after Stage 1 completion.

VI. Datasets and Experimental Design

A. Stage-Wise Dataset Allocation

Table II summarizes the planned datasets per stage, drawing on standard dermatoscopic collections and clinical text corpora [14]–[17], [20], [24], [25], [34]–[36].

B. Data Curation and Preprocessing

Because datasets originate from different centers with varying acquisition protocols, we will:

- Harmonize label taxonomies (e.g., mapping to a common ontology such as Derm1M's hierarchy [20]).
- Standardize image resolutions and color profiles.
- Create clean splits where patient-level leakage is minimized.
- For VQA, generate dermatology-specific question–answer pairs from Derm1M captions and literature where necessary, following methods that synthesize QA from medical texts [54].

VII. Implementation Plan and Status

A. Model Choices and Deployment Status

For each stage, we selected models based on performance, feasibility, and compatibility with available hardware (GPUs with $\sim$ 12 GB VRAM). The following table summarizes implementation status:

B. Training Strategy

Each stage trained separately with stage-specific optimizations: SLS (Dice+CE, LR 10, 100 ep), HC (AsymmetricFocalLoss, differential LR, 80 ep), MedProc (5×10, 10 ep on MIMIC). All employ AMP mixed precision; best validation checkpoints restored at evaluation.

C. Planned Infrastructure (Stage 8–9)

Inference via FastAPI with GPU acceleration, asynchronous task queue (Celery+Redis), Docker containerization, structured logging, and audit trails for clinical governance. Stage 8 introduces conversational reasoning via LLM + RAG, while Stage 9 focuses on deployment and monitoring.

TABLE I: Comprehensive Preprocessing Pipeline: Implemented vs. Planned Operations Across Stages

Stage	Input	Implemented Preprocessing	Planned/Future Work
Stage 1 **(Centralized)**	Raw inputs (JPEG/PNG/TIFF, PDF, TXT)	Image format validation, basic metadata extraction	Image denoising (noise2noise), PDF parsing with layout preservation, rule-based text expansion (abbreviation expansion), grammar/spell correction Language detection and multilingual support
Stage 2 **(SLS)**	Stage 1 output (768×768 RGB)	Resize to 512×512 (bilinear), normalize to ImageNet statistics, optional hair removal (morphological), detect blur/quality issues	Contrast-limited adaptive histogram equalization (CLAHE), dermatoscopic artifact removal (bubbles, ruler marks), quality-aware adaptive preprocessing
Stage 3 **(SLRC)**	SLS mask + original image	Connected components labeling, minimal bounding box extraction, box padding (10%), resize cropped ROI to 224×224 with aspect ratio preservation	Contour refinement (morphological closing), lesion-aware padding adjustment, perspective correction for oblique imaging
Stage 4 **(HC)**	Stage 3 output (224×224 RGB)	Normalize to ImageNet stats, random flips/rotations (train only), MixUp/CutMix augmentation, brightness/ contrast jitter	Instance normalization, learnable affine transformation, advanced augmentation (GridMask, AutoAugment), test-time augmentation (TTA) ensemble optimization
Stage 5 **(MedProc)**	Raw text + PDF	PDF OCR (Tesseract), text cleanup (boilerplate removal, case-folding, special char removal, spell correction), BERT WordPiece tokenization, truncate/chunk to 512 tokens (stride=100 for long docs)	Coreference resolution, clinical entity linking to ontologies (SNOMED CT), temporal relation extraction, negation handling, uncertainty quantification
Stage 6 **(IT-Fusion)** **(In Progress)**	HC output + MedProc JSON	Feature normalization (z-score), categorical embedding, text tokenization for fusion transformer	Attention mechanism weights visualization, semantic similarity calibration, feature importance ranking
Stage 7 **(VQA)** **(In Progress)**	IT-Fusion JSON + ROI image + question text	Prompt construction from JSON (convert ABCDE/E to textual format), ROI grayscale/color channel selection, question tokenization	Question understanding (intent classification), answer post-processing (hallucination filtering), grounding verification, confidence calibration

TABLE II: Planned datasets for each stage in NutriDermAI. Counts are approximate and may be refined during implementation.

Stage	Task	Dataset(s)	Train	Val	Test	Total
SLS	Segmentation	ISIC 2016, ISIC 2017 – Segmentation tasks	2,900	400	979	4,279
SLRC	ROI Cropping	ROIs derived from SLS outputs	2,900	400	979	4,279
HC (L1)	Benign vs. malignant	ISIC 2018 classification (HAM10000-style) [15]	7,000	1,015	2,000	10,015
HC (multi)	Extended classification	ISIC 2016–2024, BCN20000, MiLK-10K [14], [16], [17]	≈ 100,000 images split 70/10/20			≈ 100,000
MedProc	Clinical NLP	MIMIC-III, MIMIC-IV Note [24], [25]	50,000	5,000	10,000	65,000
IT-Fusion	Multimodal	Derm1M + HAM10000 + PAD-UFES-20 + ISIC 2016–2024 + BCN20000 + MIMIC text [14]–[16], [20]	700,000	100,000	200,000	1,000,000
VQA	Med & derm VQA	DermaVQA, SkinCancerVQA, SkinQA, PathVQA, VQA-RAD, HEALMQA [34]–[37]	80,000	10,000	15,000	105,000

TABLE III: Stage-Wise Implementation Status and Model Selection

Stage	Task	Status	Model(s)
2	SLS	Complete	ARC-UNet with optional ScaleFusionNet, ESC-UNet alternatives
3	SLRC	Implemented	OpenCV connected components + resize (224×224)
4	HC	Complete	EfficientFormerV2-S2 with hierarchical dual-head classification
5	MedProc	Complete	BioBERT fine-tuned on MIMIC-III/IV clinical text
6	IT-Fusion	In Prog.	MARIA-style multimodal transformer with feature alignment
7	VQA	In Prog.	R-LLaVA-based ROI-aware medical VQA with knowledge ensemble
8	Frontend-Backend	In Prog.	React UI + FastAPI orchestration with Docker containerization
9	Testing & Deploy	In Prog.	Integration tests, CI/CD pipeline, access control, monitoring

VIII. Experimental Results

A. Stage-Wise Performance and Algorithm Details

We present results for the completed and tested stages of the NutriDermAI pipeline. For each stage, we document the dataset used, algorithmic modifications from standard baselines, training configuration, and measured performance metrics.

1) Stage 2: Skin Lesion Segmentation (SLS) – ARC-UNet: **Dataset:** ISIC 2016-2017 (4,279 images: 2,900 train, 400 val, 979 test).

Algorithm: ARC-UNet with SE attention, residual connections, asymmetric encoder-decoder.

Training: Hybrid Dice+CE loss (0.6/0.4), Adam LR 10, batch 8, 100 epochs, L =10, early stopping patience 10, augmentation (flips, rotations ±15°, elastic deformations).

Results:

ARC-UNet achieves F1 0.8815 boundary precision with balanced sensitivity/specificity. Single-pass inference recommended for clinical deployment.

2) Stage 3: Skin Lesion ROI Cropping (SLRC) – Connected Components + Resize: **Dataset:** Derived from ISIC 2016/2017 masks (not separate dataset; uses SLS output masks on same 4,279 images).

Algorithm: Deterministic image processing pipeline:

TABLE IV: Stage 2 (SLS) - ARC-UNet Segmentation: Single-Pass vs. Test-Time Augmentation (7-view ensemble).

Metric	Single-Pass	TTA (7-view)	Improvement
Dice Coefficient	0.8506	0.8512	+0.07%
IoU (Intersection-over-Union)	0.7718	0.7726	+0.10%
Pixel Accuracy	93.70%	93.75%	+0.05%
Boundary F1-Score	0.8801	0.8815	+0.16%
Sensitivity (Recall)	0.9102	0.9118	+0.18%
Specificity	0.9756	0.9761	+0.05%
Optimal Probability Threshold	0.44		(well-calibrated)

TABLE V: Stage 3 (SLRC) - ROI Cropping Accuracy: Lesion Coverage and Border Preservation.

Metric	Value
Mean Lesion Coverage (% of lesion pixels retained in ROI)	98.4% (std: ±1.2%)
Border Preservation (Contour pixels within ROI margin)	96.7% (std: ±2.1%)
Mean ROI Size Variation *(deviation from anatomical diameter)*	0.34 mm (IQR: 0.18–0.65 mm)
Computational Time per ROI (on CPU; OpenCV 4.5.2)	12.3 ms

1) *Binary morphological cleanup:* Apply closing (dilation then erosion with disk structuring element $r = 3$) to fill small holes in segmentation mask.
2) *Connected components labeling:* Extract all distinct connected regions using 8-connectivity; keep largest component by pixel count (lesion region).
3) *Bounding box extraction:* Compute minimal axis-aligned bounding box around lesion; pad by 10% on all sides (margin preservation).
4) *Aspect-ratio-preserving resize:* Crop padded ROI; resize to 224×224 via bilinear interpolation with aspect ratio maintained (pad with black borders if needed).

Results:

High coverage (98.4%) and border preservation (96.7%) confirm reliable ROI extraction with 12.3 ms latency.

3) Stage 4: Hierarchical Classification (HC) – EfficientFormerV2-S2 + ABCD Features: **Dataset:** ISIC 2018+HAM10000+ISIC 2019-2024, 50,000 balanced samples (70/10/20 split), no patient overlap.

Algorithm: EfficientFormerV2-S2 with dual-head classification (benign/malignant + 7-class), parallel ABCD sub-pipelines.

Training: AsymmetricFocalLoss ($\gamma = 2.0$), OneCycleLR (head: 10^3, backbone: 10), batch 16×2 accumulation, 80 epochs, MixUp/CutMix augmentation, WeightedRandomSampler.

Results:

Explicit ABCD features improve interpretability and performance. TTA provides 1% improvement for critical cases; single-pass acceptable for routine triage.

4) Stage 5: Medical Information Processing (MedProc) – BioBERT Clinical NLP: **Dataset:** MIMIC-III/IV (50K train, 5K val, 10K test), no patient overlap.

TABLE VI: Stage 4 (HC) - Hierarchical Classification with ABCD Features: Single-Pass vs. TTA Ensemble (7-view).

Main Task (Benign/Malignant)	Single-Pass	TTA (7-view)	Improvement
Accuracy	94.7%	95.3%	+0.6%
Sensitivity (Recall)	0.9512	0.9618	+1.06%
Specificity	0.9387	0.9425	+0.38%
AUC-ROC	0.9751	0.9803	+0.52%
Sub-Task (7-Class Multi-Class)	**Single-Pass**	**TTA (7-view)**	**Improvement**
Accuracy	89.2%	90.1%	+0.9%
Weighted F1-Score	0.8876	0.8979	+1.03%
Macro F1-Score	0.8654	0.8762	+1.08%
Per-Class Balanced Accuracy	0.8905	0.8998	+0.93%

Algorithm: BioBERT-base + 2 transformer blocks for clinical NER, relation extraction, and rule-based Evolution (E) scoring.

Training: AdamW (LR 5×10), batch 32, 10 epochs, token limit 512 with stride 100 chunking, dropout 0.1, label smoothing 0.1.

Results:

TABLE VII: Stage 5 (MedProc) - Clinical NLP: Entity Recognition, Relation Extraction, and Evolution Scoring.

Clinical NER (Named Entity Recognition)	F1-Score
Symptoms (itching, bleeding, pain, color change, etc.)	0.8742 (Precision: 0.8901, Recall: 0.859
Body sites (site, location, distribution)	0.8934 (Precision: 0.9087, Recall: 0.879
Durations and temporal mentions	0.8156 (Precision: 0.8312, Recall: 0.801
Severity indicators (escalation phrases)	0.7821 (Precision: 0.7956, Recall: 0.769
Relation Extraction (Symptom-to-Evolution)	
AUC-ROC (symptom progression detected)	0.8934
Precision (of relations recovered)	0.8764
Recall (of ground-truth relations)	0.8523
Evolution (E) Score Quality	
Calibration: Expected Calibration Error (well-calibrated; $\sigma \approx 0.020$)	0.0201
Mean Absolute Error vs. clinician ratings (on scale 0–1; ±0.154 MAE)	0.1547

BioBERT achieves F1 0.87–0.89 on clinical entity extraction with well-calibrated uncertainty (ECE 0.0201) for downstream fusion.

B. Integration and Data Flow Validation

Early integration tests across Stages 2–5 confirm:

- *Data flow:* SLS mask (shape: 512×512 binary) → SLRC ROI (224×224 RGB) → HC classification (logits + ABCD features) + MedProc JSON (symptoms + E-score).
- *Compatibility:* Structured outputs (JSON for MedProc, numpy arrays for image-based stages) properly serialized and deserialized between stages.
- *Computational efficiency:* Per-sample latency (all 4 stages sequentially on NVIDIA A100): SLS 280 ms, SLRC 12 ms,

HC 45 ms, MedProc 180 ms; total 517 ms for image+text input (acceptable for clinical batch processing).

IX. Evaluation Plan

A. Metrics and Validation

Each stage is evaluated independently on standard benchmarks (Tables IV–VII):

HC: Accuracy and AUC-ROC for benign/malignant and multi-class tasks. [**Completed**: 94.7% main-task, 89.2% multi-class accuracy]

MedProc: F1 for clinical entity extraction and relation classification. [**Completed**: Calibration ECE = 0.0201]

B. Future End-to-End Evaluation

Planned validation includes: (1) comparative analysis of image-only vs. multimodal performance (2) user studies with dermatologists, and (3) fairness analysis across skin tones.

X. Component Validation and Design Benefits

The modular four-stage architecture enables independent evaluation and reusability:

1) **Independent evaluation:** ISIC benchmarks for SLS/HC; MIMIC for MedProc; deterministic validation for SLRC.
2) **Reusability:** SLS and HC deployable independently; optional MedProc integration for comprehensive assessment.
3) **Structured outputs:** JSON (text), numpy arrays (images), and calibrated confidence scores enable explainability and audit trails.
4) **Scalability:** Microservice-compatible design supporting containerized cloud/on-premise deployment.

XI. Implementation Status, Limitations, and Future Work

A. Current Scope and Completed Work

Four core modules fully implemented and evaluated on standard benchmarks:

- **Stage 2 (SLS):** Dice 0.8506, IoU 0.7718, 93.70% accuracy (Table IV).
- **Stage 3 (SLRC):** 98.4% coverage, 96.7% border preservation, 12.3 ms latency (Table V).
- **Stage 4 (HC):** 94.7% main-task, 89.2% multi-class (single-pass); TTA 95.3% and 90.1% (Table VI).
- **Stage 5 (MedProc):** NER F1: 0.8742 (symptoms)–0.8934 (body sites); ECE 0.0201 (Table VII).

B. Pending Stages (1, 6–9)

Remaining stages require implementation: Preprocessing (image/text normalization); IT-Fusion (multimodal transformer); VQA (R-LLaVA); Stage 8 LLM+RAG integration; Frontend/backend and deployment; clinical validation with dermatologists (n = 10–20); fairness analysis across skin tones; regulatory documentation.

C. Design Assumptions

Sequential processing; ABCD computed from image features (E from clinical text); stateless inference; English-language text.

D. Safety and Ethical Considerations

Designed as decision support (not standalone diagnostic); escalation when confidence is low; secure data processing; periodic fairness monitoring across skin tones [33], [43].

XII. Conclusion and Future Work

This paper presents NutriDermAI, a four-stage working implementation of an ABCDE-aware multimodal dermatology assistant: skin lesion segmentation (Dice 0.8506), ROI extraction (98.4% coverage), hierarchical classification with explicit ABCD metrics (94.7% accuracy), and clinical NLP for evolution scoring (F1 0.87–0.89, ECE 0.0201). The modular architecture with structured JSON outputs enables interpretability, independent evaluation, and redeployiyibility suitable for clinical governance and future deployment.

A. Completed Core Modules

Four stages fully implemented and evaluated (details in Tables IV–VII): SLS (Dice 0.8506), SLRC (98.4% coverage), HC (94.7% accuracy), and MedProc (F1 0.87–0.89 with ECE 0.0201 calibration). End-to-end integration tested with 517 ms total latency.

B. Pending Stages (1, 6–9)

Remaining stages require implementation: Preprocessing (image/text normalization); IT-Fusion (multimodal transformer); VQA (R-LLaVA); Frontend/backend and deployment; clinical validation with dermatologists (n = 10–20); fairness analysis across skin tones; regulatory documentation.

C. Significance and Impact

NutriDermAI demonstrates feasibility of explainable, ABCDE-aware dermatology support through four validated core stages: reliable segmentation (Dice 0.8506, F1 0.8815), efficient ROI extraction (98.4% coverage, 12.3 ms), high-accuracy classification (94.7% benign/malignant), and robust clinical NLP (F1 0.87–0.89, ECE 0.0201). Modular design with structured JSON outputs enables interpretable, auditable, and redeployable components suitable for clinical governance.

D. Next Steps and Future Outlook

Prioritize Stages 1, 6–7 implementation with clinical validation studies. NutriDermAI provides a research-validated foundation for clinical ABCDE-aware dermatology systems combining interpretability, strong empirical performance, and modular reusability.

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